

Siddhi Kiran Bajracharya

Mountain View, CA | siddhikiran.bajracharya@gmail.com | 605-728-6309 | siddhibajracharya.com.np
linkedin.com/in/siddhikiran | github.com/siddhi47

Education

University of South Dakota, MS in Computer Science

Aug 2022 – Dec 2023

- GPA: 4.0/4.0
- **Coursework:** Computer vision, Machine learning and pattern recognition

Experience

Senior Machine Learning Engineer, ONRAMP – New York

Sep 2025 – Present

- Architecting the ML/AI infrastructure for a high-impact payments system within the U.S. freight industry, a sector crucial to the national supply chain.
- Designing scalable and efficient backend services (utilizing GCP and Kubernetes) for new AI-powered features, ensuring the underlying architecture supports the economic sustainability of the platform over legacy systems.
- Providing technical leadership in the initial design and specification of optimization and recommendation algorithms that will deliver efficiency gains for carriers and merchants..
- Applying deep expertise in efficient algorithm design and data-intensive systems to build a robust, high-performance solution that advances technological innovation in the traditionally outdated payments sector.

Senior Machine Learning Engineer, i8labs – Mountain View, CA

Feb 2024 – Sep 2025

- Achieved an 8% increase in mean Average Precision (mAP) by training and fine-tuning custom YOLO computer vision models using the Ultralytics framework on AWS cloud.
- Improved deployment efficiency in IoT devices by reducing the size of the Docker image by 60%.
- Enhanced model accuracy by establishing standards for tools and procedures in the internal semi-auto annotation pipeline and reduced the manual effort by 80%.
- Ensured 99% up-time of the deployed services using tools such as Grafana for logging monitoring and Balena for device management.

Machine Learning Engineer, Leapfrog Technology – Nepal

Aug 2021 – Aug 2022

- Led a team of two machine learning engineers to develop an end-to-end deep learning object detection and tracking for IoT devices using TensorRT on Jetson Nano.
- Achieved 20% velocity gain in the overall sprint cycle by defining and implementing standard machine learning practices including unit testing, containerization, and CI/CD pipeline.
- Facilitated smoother data ingestion and processing using Python and Fast API for secure data integration with IoT devices.
- Implemented MLFlow for end-to-end machine learning life-cycle management, including experiment tracking, model tracking, and model versioning.

Data Scientist, extensoData – Nepal

Sept 2018 – July 2021

- Automated micro loan disbursement in financial institutions in Nepal by 60% by applying machine learning algorithms using Tensorflow and PyTorch, integrated with tools like Python, Pentaho, shell scripts, and Apache Airflow.
- Improved data entry speed by 90% by integrating the AWS cloud platform with Tesseract and Python scripts to automate the extraction of text from national identification cards.
- Facilitated better communication with stakeholders by demonstrating expertise in all phases of the modeling pipeline using Python and R, effectively communicating complex technical ideas.

Publications

Deep Spectral Features to Detect Atrial Fibrillation using Single-Lead ECG Signals 10.1109/CAI54212.2023.00074	2023
Cracking the Machine Learning Code: Technicality or Innovation? 10.1007/978-981-97-2720-9	2024

Projects

Multimodal emotion analysis

- Designed a multimodal neural network using LLMs and visual models to identify emotions in speech.
- Tools Used: PyTorch lightning, Python.
- github.com/siddhi47/multimodal-emotion

Pyspark Recommendation

- Scalable recommendation system for user-rating data.
- Tools Used: HDFS, PySpark, AWS, MongoDB, Docker, Flask
- github.com/siddhi47/pyspark-recommendation

Certifications

MLOps Machine Learning Operations by Duke University C2229E7K8YCJ	Aug 2024
Container Orchestration using Kubernetes (Coursera) 1df59f5037a370f3e6107779451aeca2	Aug 2024
Microsoft Azure AI Fundamentals AI-900 Exam Prep by Microsoft. H3WL0WOPFW79	Jan 2025

Technologies

Languages: Python, C, C++, SQL, Bash, GO.

Frameworks and Libraries: Pandas, NumPy, scikit-learn, Tensorflow, Pytorch, Ultralytics, Django Rest, FastAPI, Flask, OpenCV, depthai, MLFlow, Ultralytics, depthai.

Database: MySQL, MongoDB, SQLite, Postgres.

Version Control: Git, Butbucket.

Cloud: AWS (EC2, S3, EKS), GKE, Vertex AI (GCP).

MLOps: MLFlow, ClearML.

Orchestration: Apache airflow, Kubernetes (AWS EKS), Helm.

Hardware: OAK Camera, NVIDIA Jetson, ESP32, RaspberryPi.

Observability: Grafana, Sentry